Supplementary Information

This supplement gives the implementation details and sensitivity results that are needed to reproduce the study but are not needed to follow the main paper. It covers the image and semantic-model roles, street-image acceptance, numerical settings, provenance, convergence, the material-evidence control, timing boundaries, and additional limitations. Unless marked as the separate fixed-grid diagnostic, numerical statements refer to the verified ten-site calculations under the first-material-interaction transport contract.

I. Image labels and material mapping

The image-labeling pass uses two models with separate roles. Mask2Former with the 65-class Mapillary Vistas vocabulary assigns one street-scene object class to every pixel. It therefore supplies the complete object map and identifies small objects such as poles, signs, curbs, and bicycle racks. One Vistas building label can contain brick, stone, render, glass, and metal. SAM 3 supplies the open-vocabulary material labels needed inside such broad object classes. The production catalog contains 60 text prompts and 61 raster identifiers including the unlabeled identifier. Examples include “brick facade”, “glass window”, “metal cladding panel”, and separate ground, grass, shrub, tree, and forest concepts. The prompts are also gated by the Vistas classes. A class must occupy at least 256 pixels in a 1536×1536 crop before its related prompts are sent to SAM 3. The gate reduces work and removes prompts that have no matching object in the view. The complete prompt list and its object, material, attribute, and vegetation mappings are stored in `config/semantic_concepts.json`.

The four rectilinear crops of each 360-degree street image face $0°$, $90°$, $180°$, and $270°$, zero pitch, a $90°$ field of view, and 1536 pixels per side. Mask2Former runs at 1536 pixels. SAM 3 uses its trained 1008-pixel input, a score threshold of 0.35, and prompt batches of 32. Production fixes the full model, source, and catalog identities listed in Table I. It refuses another model pair or a catalog change, even if the number of prompts remains the same.

Rays from each 360-degree street image intersect the original city mesh. Each observed mesh triangle has a common 8×8 barycentric material grid. Its coordinates are fixed to the triangle and do not depend on the camera. Each camera first reduces all of its rays in one grid cell to at most one confidence-weighted contribution. Camera means are then added across views. Range and raw pixel density give no extra weight. Two images can therefore place a material boundary at different positions on one large mesh triangle. Both observations are accumulated in the same barycentric cells and remain a distribution. The transport lookup uses the original triangle identifier and the barycentric coordinates of each ray hit. The highly tessellated mesh shown for inspection is a display object and is not the transport mesh.

Material labels remain probabilistic when observations are combined. Vistas-backed pixels use the declared full $p(\text{material} \mid \text{entity})$ table. Concept-backed pixels use the full material distribution of the detected prompt. Transport removes probability assigned to air, unknown material, people, vehicles, and participating volumes. It assigns an image-derived structural material only when the remaining compatible structural probability is strictly greater than 0.5. An exact tie and any unsupported material-map cell use the geometric face material. Reflected power uses the posterior-weighted material coefficients. The specular sampling probability uses the posterior-weighted reflected specular share. Grass keeps the geometric ground interface. Woody canopy labels are nonblocking until a closed canopy volume can supply path lengths for volume attenuation.

The per-image inspection mesh is a separate audit product. It cuts visible city-mesh triangles at object boundaries and stores each accepted piece with its source triangle, source image, confidence, area, and visible fraction. A parallel table keeps rejected pieces when a reliable inverse projection exists. Reason codes include a degenerate or subpixel triangle, clipping outside the crop, occlusion by the city mesh, clutter in front, a transient object, missing object labels, area below the retained minimum, a grazing plane, a pixel that does not belong to the city mesh, and a mesh or camera-pose conflict. A rejected piece can lie on the city mesh because the code describes why that image-space piece did not receive an accepted object or material label. Rejected pieces do not replace or remove the original transport geometry.

A. *Material labels on reached surfaces*

The audit replays all ten verified routes and 64 seeds with the production geometry, source curve, material map, body, 200,000 primary rays, and 4,096 output cells. It classifies the exact reflection point of every retained order-1 specular path and the first blocking surface of every accepted first-diffuse event. Category counts, transport contribution, body-coupled fields, and the complete replay all match the verified arrays. Direct transport is not assigned a category because it has no material interaction.

Image-derived materials inform most of the pooled non-direct body contribution because the exact specular term is both larger and more often image-mapped. The first-diffuse component reaches different surfaces. Most of its body-coupled contribution reaches geometry with no image label or with a label rejected by the object-material compatibility test. The nonblocking woody category is zero for retained terminal interactions because a nonblocking crossing cannot be the first blocking surface. This zero does not imply that woody canopy evidence is absent from the routes.

II. Street-image alignment and acceptance

The alignment matches the segmented sky boundary in a 360-degree street image to the upper edge of the city mesh. The stored residual is the angular skyline error. An accepted camera pose has a residual no greater than $4°$. The second test casts the directions labeled as sky from the fitted camera into the mesh. It records the fraction that hits geometry and the median range of those hits. A camera is classified as inside the geometry when the hit fraction is greater than 0.5 and the median range is less than 2 m. The paired condition matters because a camera inside a wall can still obtain a small silhouette

TABLE I
FIXED MODEL IDENTITIES USED BY THE PRODUCTION MATERIAL MAP.

Item	Immutable identifier
Mask2Former weights	`4772b6bf101d91f2534c106dc524d906aeb3c68a`
SAM 3 weights	`3c879f39826c281e95690f02c7821c4de09afae7`
SAM 3 source	`96914d2425f90a64f45ca977c2b5165418099543`
Parsed concept catalog	`66d0dfefba87bde5081cfa82108ca60d47c79cf641df3ec44129ec20bb90453b`

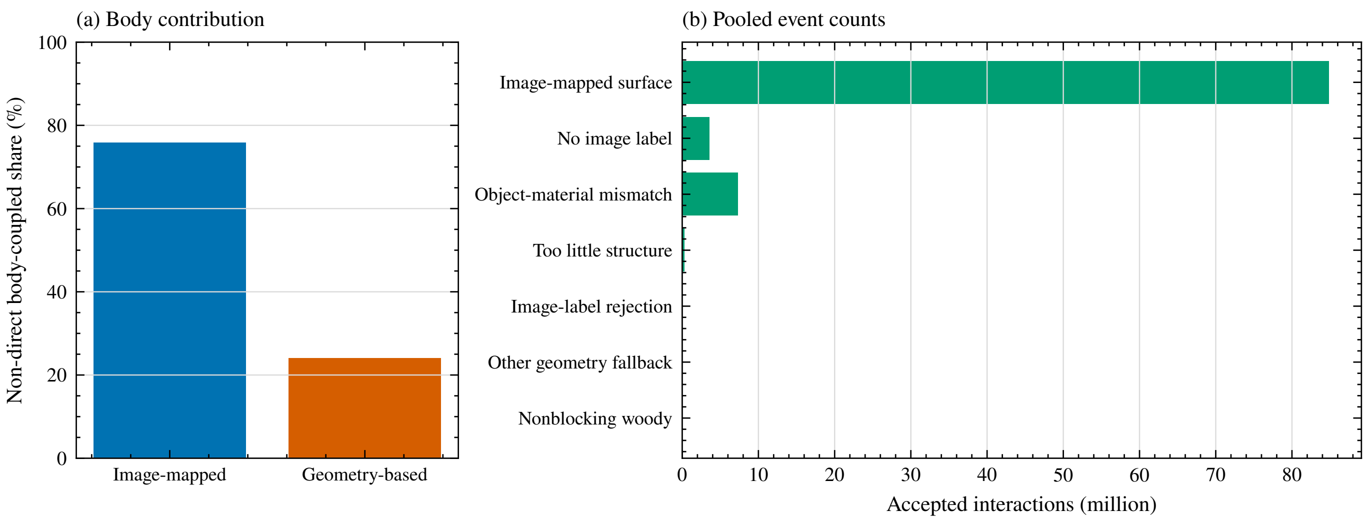

Fig. 1. Material labels on surfaces reached by modeled paths. Panel (a) partitions the pooled non-direct body-coupled whole-body SAR contribution between image-mapped and geometry-based materials. Panel (b) gives accepted retained interaction counts for all seven audit categories. Direct transport is not applicable because it has no material interaction.

TABLE II
SHARE OF POOLED BODY-COUPLED WHOLE-BODY SAR WITHIN EACH RETAINED NON-DIRECT COMPONENT. OTHER FALLBACK COMBINES INSUFFICIENT STRUCTURAL PROBABILITY, A REJECTED IMAGE LABEL, AND THE REMAINING GEOMETRY-BASED CATEGORY.

Component	Image mapped	No image label	Object-material mismatch	Other geometry fallback
Order-1 specular	80.643%	8.774%	10.530%	0.052%
First diffuse	13.337%	44.345%	42.314%	0.004%
Combined non-direct	75.903%	11.280%	12.769%	0.049%

residual. A large conflict at long range is recorded as a separate low-quality state.

The current acceptance policy also requires complete sky checks and a best-fit camera height inside the tested altitude interval. It rejects a pose with a missing residual, a residual above 4°, missing sky diagnostics, the near inside-geometry signature, or an optimum at the altitude bound. Each material map stores the policy version that selected its cameras. New builds use `registration-admission-v2`. The verified Korenmarkt map uses its recorded version-1 policy. An accepted camera can still be skipped if its dense and SAM label images are missing, incomplete, malformed, or inconsistent with the fixed vocabulary. These rejections prevent an image, pose, model, or mesh mismatch from entering the combined material map.

III. NUMERICAL SETTINGS AND FILE VERIFICATION

The receiver is $1.5\,\mathrm{m}$ above the city surface. Route points are placed at approximately $6\,\mathrm{m}$ intervals along the route, with aligned street-image positions retained as route anchors. The visible-roofline calculation finds boundary and crease edges that separate the first visible mesh hit from sky. Duplicate mesh-edge contacts are combined. The resulting three-dimensional edge lengths are the numerical weights. Each possible transmitter position is shifted vertically by $0.5\,\mathrm{m}$ to avoid self-occlusion. Table IV gives the retained counts.

The first-diffuse term uses a reciprocal next-event estimator. For each primary direction $\mathbf{u}_n$ drawn uniformly over 4π, the tracer keeps the first blocking surface. It then draws one roofline segment from the arc-length weights and tests the connecting segment. If the connection is visible, its unscaled contribution is

$$w_n = \frac{R_n(1-\kappa_n)}{\pi}\,\frac{[\mathbf{n}_n\cdot\mathbf{d}_n]_+}{r_n^2}. \qquad (1)$$

Here R_n is the unpolarized Fresnel power reflectance, κ_n is the coherent specular share, $\mathbf{d}_n$ points from the surface to the sampled transmitter position, and r_n is the connecting distance. Occluded connections contribute zero. The estimate is $(4\pi/N)\sum_n w_n$. The launch direction determines the physical arrival direction by reciprocity. Only these sampled first-diffuse contributions are accumulated in the 4,096-cell Fibonacci grid. Direct sources and accepted one-reflection specular paths retain their exact arrival directions.

TABLE III
CONFIGURATION OF THE VERIFIED TEN-SITE RESULT.

Item	Production value
Sites and routes	Brussels, Ghent, Krakow, London, Madrid, Mexico City, Milan, Prague, Tokyo Hachiko, and Toulouse. The fixed routes contain 14, 10, 16, 22, 14, 11, 23, 22, 16, and 15 route points (163 total).
Geometry and frequency	Original photogrammetric city mesh within a 250 m horizontal radius at 15 GHz. The circular crop area is $196{,}349.54\,\mathrm{m}^2$.
Transmitter model	Roofline visible from the route. Areal density sets the expected number of transmitters. Physical three-dimensional roofline length assigns their relative probability to each segment.
Exposure normalization	Per unit $\rho_A P_{\mathrm{EIRP}}$. Normalized whole-body SAR has unit $\mathrm{m}^2\,\mathrm{kg}^{-1}$.
Transport	Exact direct term, exact one-reflection specular term, and stochastic next-event estimation of one diffuse reflection at the first blocking surface.
Surface model	Image-derived material map with measured finish roughness. The material probabilities are evaluated at the exact hit point.
Monte Carlo	200,000 IID primary rays per route point and replica. Seeds 7 through 70 give 64 replicas. Stored convergence results use 16, 24, 32, 48, and 64 replicas.
Angular output	4,096 passive cells collect only the first-diffuse estimate. Exact direct and one-reflection specular paths retain their arrival directions. The cells do not control launch directions.
Body	Duke with 56,024 surface elements, area $1.87250\,\mathrm{m}^2$, mass 72.4 kg, and $T_0 = 0.500140$. The phantom faces along the walk. Body coupling uses the level-2 surface-field model.

TABLE IV
NUMBER OF ROUTE POINTS AND ROOFLINE SEGMENTS.

Site	Route points	Roofline segments	Length (m)
Brussels	14	450	158.53
Ghent	10	457	157.54
Krakow	16	320	94.97
London	22	635	296.67
Madrid	14	207	79.40
Mexico City	11	164	63.69
Milan	23	502	437.63
Prague	22	502	267.15
Tokyo Hachiko	16	400	248.76
Toulouse	15	283	72.10

For incidence cosine c, complex relative permittivity ϵ_r, surface root $q = \sqrt{\epsilon_r - (1-c^2)}$, and wavelength λ, the production law is

$$R = \tfrac{1}{2}\left(\left|\frac{c-q}{c+q}\right|^2 + \left|\frac{\epsilon_r c - q}{\epsilon_r c + q}\right|^2\right), \tag{2}$$

$$\kappa = \exp\left[-\left(\frac{4\pi s c}{\lambda}\right)^2\right]. \tag{3}$$

Thus the reflected specular and diffuse shares are $R\kappa$ and $R(1-\kappa)$. The surface-finish roughness s excludes larger periodic relief. Table V gives the evaluated 15 GHz material table. Metal uses the finite-conductivity value in the same complex-permittivity convention.

The deterministic order-1 search uses a conservative mirrored-receiver triangle-cone broad phase when the candidate set reaches 20,000. Its CUDA Float64 implementation leaves the host exact final test unchanged. The normal adaptive candidate budget is 120 million. One Tokyo route point has zero accepted order-1 paths, so a relative stopping rule cannot close around zero. That case records and fully enumerates a 320 million candidate cap. Body coupling uses fixed blocks of 512 directions. Candidate caps, broad-phase thresholds, chunks, and block sizes control computation. They do not change the stated physical model.

The result contains 163 route points, 10,432 point-replica fields, 640 site-replica runs, and 2.086 billion stochastic primary rays. Every campaign manifest lists 42 files. All 420 recorded file hashes pass. The aggregate JSON, CSV, PDF, and PNG also match their artifact manifest. Across the 10,432 fields, the maximum raw residual between the saved total and the sum of direct, all-specular, and first-diffuse components is $1.214 \times 10^{-17}\,\mathrm{m}^{-2}$. The CUDA body result agrees with the NumPy reference to a maximum relative error of 6.64×10^{-16} in the verified benchmark. The verified artifacts are stored under the roofline campaign output package named `ten_city_route_production64_v1`. The Duke STL has SHA-256 `781e65ef3882f134 7669e0ddca5dafa82cd6368dddd6b9e80 1dc49613822fe3b`. The verified body-area array has SHA-256 `5dd410b9512fdb0274dd1ac1fb8f7b70 87564abfed998d57a25065e87ffebd6a`. Each calculation manifest also verifies the city mesh, material map, roofline segments, source weights, route arrays, material tables, and executable configuration.

IV. REPLICA CONVERGENCE AND LOWER TAILS

The 64-replica extension uses seeds 7 through 70 and nested looks of 16, 24, 32, 48, and 64. Its first 16 replicas exactly match the verified campaign arrays after timing fields are removed. Mexico City route points 0, 1, and 3 and Tokyo Hachiko route points 13, 14, and 15 remain the six shadowed points. The first-diffuse estimate is their only nonzero modeled contribution. From 48 to 64 replicas, their largest pointwise whole-body SAR changes are 0.0125 and 0.0104 dB. The route q_{10} changes are 0.00344 and 0.00491 dB, and their whole-replica bootstrap widths are 0.364 and 0.0538 dB.

The bootstrap resamples complete replicas jointly over all route points, so it preserves spatial dependence within one replica. Its 2,000 draws use PCG64 with the authenticated analysis seed 20260814. All ten identities, manifests, component closures, and common inputs pass, and both lower tails meet the stated 48-to-64 aggregate criteria. Mexico City nevertheless retains rare-event first-diffuse behavior: its maximum positive

TABLE V
EVALUATED TRANSPORT MATERIALS AT 15 GHZ. THE ROUGHNESS COLUMN IS RMS HEIGHT.

Class	$\Re\epsilon_r$	$-\Im\epsilon_r$	s (μm)	Class	$\Re\epsilon_r$	$-\Im\epsilon_r$	s (μm)
Asphalt concrete	4.83	0.569	450	Metal	1.00	1.20×10^7	5
Brick	3.91	0.044	30	Plasterboard	2.73	0.130	150
Ceramic	7.07	0.081	280	Plywood	2.71	0.396	50
Chipboard	2.58	0.215	50	Polymer	1.99	0.103	50
Concrete	5.24	0.461	600	Soil	5.24	0.461	600
Fabric	1.99	0.103	50	Water	5.24	0.461	600
Glass	6.31	0.162	1	Wood	1.99	0.103	50
Marble	7.07	0.081	280				

TABLE VI
CONVERGENCE FROM 48 TO 64 REPLICAS. BOOTSTRAP WIDTH IS THE 95% INTERVAL WIDTH FOR ROUTE q_{10} WHOLE-BODY SAR. SHADOW CHANGE IS THE LARGEST STEPWISE CHANGE AMONG THE SIX SHADOWED ROUTE POINTS IN MEXICO CITY AND TOKYO HACHIKO.

Site	q_{10} change (dB)	Bootstrap width (dB)	Shadow change (dB)
Brussels	0.000114	0.000592	–
Ghent	0.0000723	0.000221	–
Krakow	0.000130	0.00107	–
London	0.0000427	0.000671	–
Madrid	0.00000615	0.000374	–
Mexico City	0.00344	0.355	0.0125
Milan	0.0000883	0.000236	–
Prague	0.0000203	0.000222	–
Tokyo Hachiko	0.00491	0.0528	0.0104
Toulouse	0.0000110	0.000173	–

replica contribution is 5738 times its positive-replica median, compared with 2.38 for Tokyo Hachiko. These intervals apply to each fixed route. They do not include route-selection or city-sampling uncertainty.

V. PRIMARY-RAY AND ANGULAR-CELL BUDGETS

The budget test used paired seeds and common random numbers at the Madrid, Mexico City, and Prague routes. Exact direct and all-specular components were byte identical in every paired comparison, and the 200,000-ray, 4,096-cell replay matched the verified baseline. The largest route-median whole-body-SAR change among the five cheaper settings was 0.000797345 dB. The directional first-diffuse fields did not pass the stated test. The largest site q_{90} normalized-L_1 differences were 0.877498, 0.678942, and 0.400218 at 25,000, 50,000, and 100,000 rays, respectively. At 1,024 and 2,048 cells, they were 0.491265 and 0.455431. Each value exceeds the 0.1 limit. The largest absolute whole-body-SAR changes at a shadowed Mexico City point were 0.707362, 0.563841, and 0.217259 dB for the three ray settings. They were 0.00206975 and 0.00088674 dB for the two cell settings.

Ray cuts also do not give a defensible end-to-end speedup. At 25,000 rays, estimator-wall-time ratios relative to the baseline range from 0.991 to 1.064 across the three routes. Exact all-specular work takes 8.06 to 23.79 s in the baseline, whereas stochastic tracing takes 1.15 to 2.40 s. The q_{90} variance-time ratios for every ray-reduced arm exceed one, with a minimum of 3.16. The reported timing therefore leaves the exact specular stage dominant while the reduced ray settings increase first-diffuse variance. Figure 3 retains the 200,000-ray, 4,096-cell setting as the production baseline.

TABLE VII
IMAGE-TO-GEOMETRY CHANGE IN ROUTE QUANTILES OF NORMALIZED WHOLE-BODY SAR. POSITIVE VALUES MEAN THAT THE IMAGE-DERIVED RESULT IS LARGER.

Site	q10 (dB)	q50 (dB)	q90 (dB)
Madrid	0.233	0.249	0.269
Mexico City	24.84	-0.158	0.104

VI. PAIRED MATERIAL-MAP CONTROL

The control replaces the image-derived material map with geometry-based materials in Madrid and Mexico City. Each pair keeps the city mesh, route, source curve, source weights, body, frequency, transport steps, ray count, output cells, and 16 seeds fixed. A strict compatibility check permits changes only in the material mode, the material arrays and hashes, the run configuration, and the presence of the material-map files. Both paired manifests pass all 42 file hashes. The comparison therefore measures sensitivity to the image-derived material map under the current model. It does not isolate reflectance because the map also changes roughness, specular share, and the nonblocking state of woody vegetation. It does not measure semantic or material accuracy.

The direct component is identical within each pair. In Madrid, the image-derived map raises the route-median all-specular whole-body SAR by 1.89 dB and lowers the first-diffuse component by 12.43 dB. Their combined effect changes the total route median by 0.249 dB. Mexico City's 24.84 dB q10 change is set by three shadowed route points where both alternatives are near zero and first-diffuse transport is the only nonzero modeled contribution. The Mexico City median and q90 changes are -0.158 and 0.104 dB. The two sites show that component changes can be much larger than the change in total normalized whole-body SAR. The result supports a material-map sensitivity statement for these routes only.

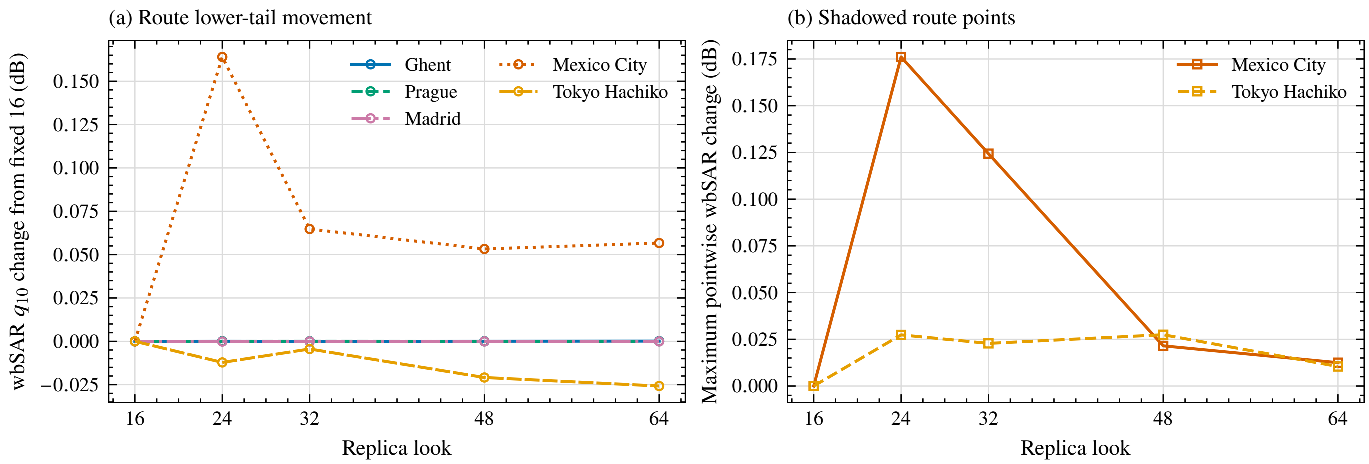

Fig. 2. Convergence through 64 replicas. Panel (a) shows route q_{10} whole-body SAR changes from the verified 16-replica value. Panel (b) shows the largest stepwise change among the three shadowed route points in each of Mexico City and Tokyo Hachiko. The final 48-to-64 changes satisfy the stated aggregate lower-tail criteria. Mexico City retains rare-event first-diffuse behavior.

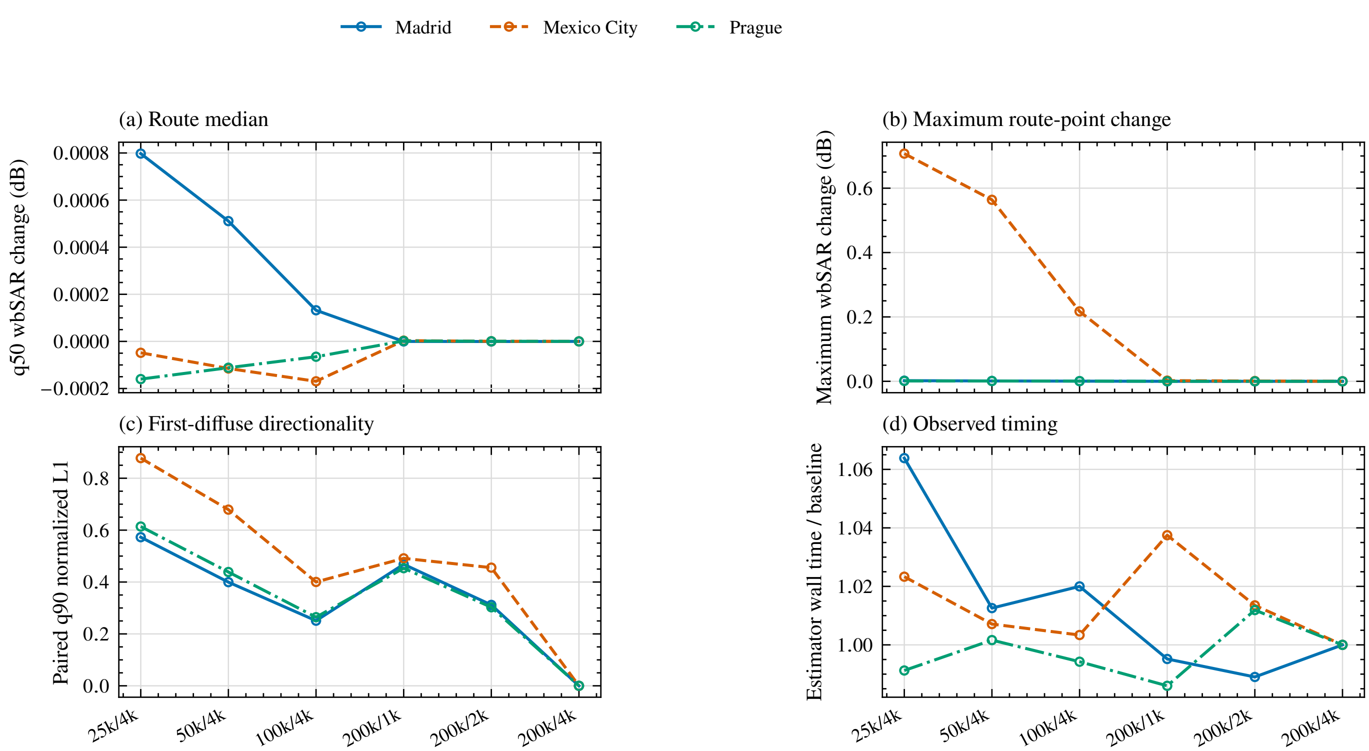

Fig. 3. Primary-ray and angular-cell budget sensitivity under the current first-material transport contract. All five cheaper settings leave route-median whole-body SAR nearly unchanged, but each exceeds the directional first-diffuse gate. Exact direct and order-1 specular components remain invariant. The diagnostic retains 200,000 rays and 4,096 cells.

VII. Separate geometric fixed-grid diagnostic

A separate diagnostic describes ten urban locations with a deterministic point design. At each location, the calculation builds a complete 6 m walkable-ground lattice within 90 m and retains 64 observation points evenly across its nearest-neighbor ordering. The visible roofline transmitter curve is built from these same 64 points. Surface properties use geometry-based priors, and the Duke phantom faces north at every point. The calculation uses the same first-material transport model with 16 seeds, 200,000 primary rays per point and seed, and 4,096 first-diffuse output cells. The fixed-grid design remains separate from the pedestrian-route calculations and is not combined with the route results.

Across the ten fixed-grid descriptions, the ratio of the largest to smallest normalized whole-body-SAR quantile is 2.02 for q_{10}, 2.01 for q_{50}, and 2.23 for q_{90}. The pooled component shares range from 75.2% to 80.6% for direct transport, 13.1% to 19.6% for order-1 specular transport, and 4.1% to 7.1% for first-diffuse transport.

Changing the design from 16 to 64 points, evaluated with the first four seeds, moved the site quantiles by at most 2.26 dB for q_{10}, 1.38 dB for q_{50}, and 3.47 dB for q_{90}. By contrast, the change from eight to 16 seeds moved them by at most 0.0036, 0.0017, and 0.0020 dB. With the 64-point transmitter curve fixed, the change from 32 to 64 receiver points was as large as 1.36, 0.219, and 0.636 dB. The calculation is therefore far more sensitive to point selection than to the number of seeds.

Each quantile describes exactly 64 fixed observation points. Its reported conditional Monte Carlo standard error is the sample standard deviation of the 16 within-seed 64-point quantiles divided by $\sqrt{16}$. This standard error is not total uncertainty. It does not include uncertainty from the grid design, body orientation, material priors, or transmitter-curve design. The values therefore describe only the declared fixed-grid calculation.

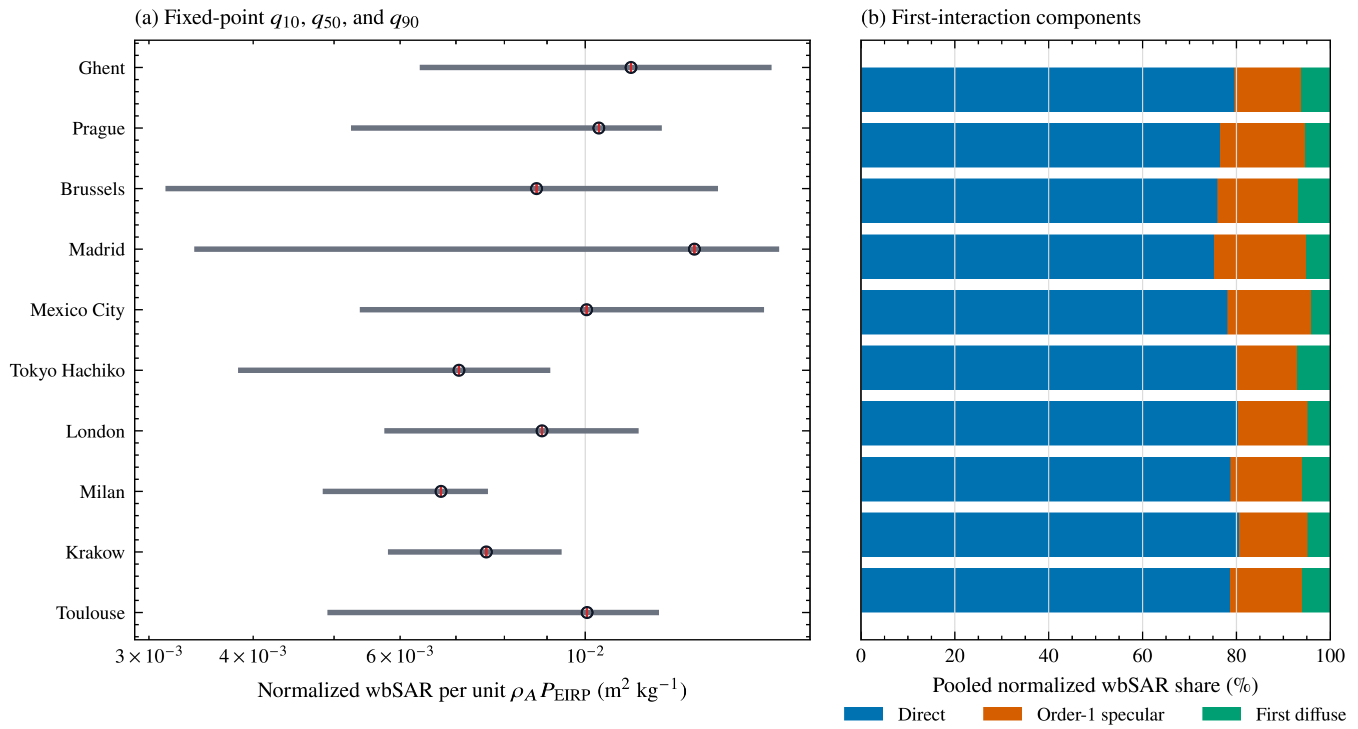

Fig. 4. Separate geometric fixed-grid diagnostic at ten urban locations. Panel (a) shows q_{10}, q_{50}, and q_{90} across the 64 fixed observation points. Red bars give the 16-seed conditional Monte Carlo standard error of q_{50}. Panel (b) shows the pooled additive component shares. These results are not combined with the route calculation.

VIII. Ten-route production detail

The ten selected routes contain 163 observation points. Every route uses image-derived surface properties, a Duke phantom facing along the walk, 64 independent seeds from 7 through 70, 200,000 primary rays per point and seed, and 4,096 first-diffuse output cells. Direct, exact order-1 specular, and first-diffuse transport use the first-material model described in the main text. The selected routes are fixed case studies, not a population sample or a city ranking.

The route-median normalized whole-body SAR ranges from $0.009016\,\mathrm{m}^2/\mathrm{kg}$ in Milan to $0.1291\,\mathrm{m}^2/\mathrm{kg}$ in Mexico City per unit $\rho_A P_{\mathrm{EIRP}}$, a factor of 14.3. Fig. 5 shows the full route ranges. The low q_{10} values in Mexico City and Tokyo Hachiko come from the shadowed points described in Section IV, rather than a shift in the central route values.

Direct transport accounts for 69.8% to 89.5% of route-summed absorbed power. The order-1 specular share ranges from 9.1% to 29.9%, and the first-diffuse share ranges from 0.301% to 5.02%. Between 48 and 64 replicas, the largest absolute route-quantile changes across the ten routes are 0.00491 dB for q_{10}, 0.000157 dB for q_{50}, and 0.0000762 dB for q_{90}. The ten routes contain 10,432 point-replica body fields and 2.0864 billion primary rays.

IX. Computation time

The repeated numerical campaigns start from a ready city scene. Their wall times on one A6000 are 29.79 s for Korenmarkt, 69.02 s for Prague, 40.70 s for Madrid, 30.92 s for Mexico City, and 50.11 s for Tokyo Hachiko. These times include the 16 transport replicas and body coupling for the complete fixed route. They exclude image and mesh acquisition, image-to-mesh alignment, image labeling, depth processing, and material-map construction. Persistent transport caches and exact conservative broad-phase screening reduce repeated work with no change to the saved scientific arrays.

The full scene-preparation path has no complete timing record. Available 14-image records give 21.3 to 26.6 min for the hybrid semantic pass and 5.3 to 13.6 min for material-map construction. Acquisition, alignment, depth, and combination were not all timed separately. One live A6000 parity check took 116.981 s for one image with independent model sessions. A shared session processed two images in 199.228 s and reproduced every scientific image array, concept cache array, prompt gate, and normalized metadata field exactly. These values are implementation checks. They do not define a universal time per image. A single cold end-to-end time is therefore not reported.

X. Additional limitations

The transport result contains exact direct transport, exact one-reflection specular transport, and one diffuse reflection. A sampled first-diffuse path stops at that reflection. The model omits a specular reflection after the diffuse event, further specular reflections, and all other multipath. The controlled open-square test validates the first-diffuse normalization, visibility, inverse-square loss, and cosine factors. The maximum adjoint-versus-Sionna difference is 0.0621 dB for the bounced term and 0.0344 dB for total transport in that test. The same three-way

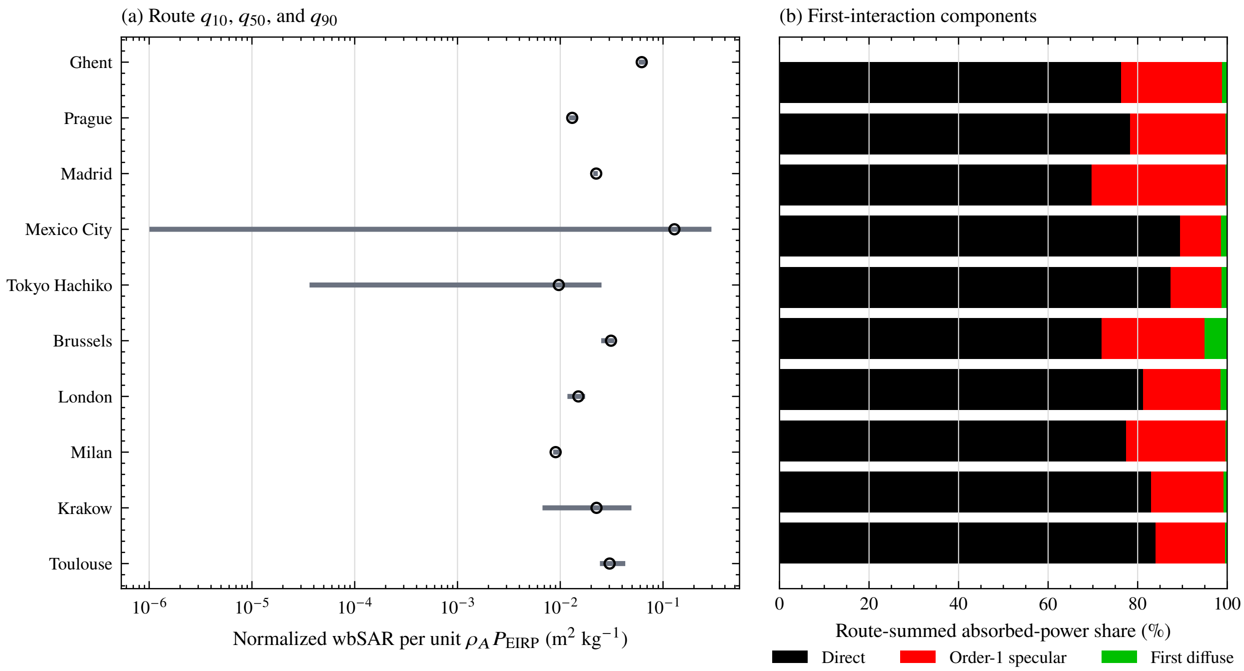

Fig. 5. Per-route exposure and component detail. Panel (a) shows the route q_{10}–q_{90} interval and an open marker at q_{50} for normalized whole-body SAR. Panel (b) shows the route-summed additive absorbed-power shares. All routes use 64 replicas and image-derived surface properties. The selected routes do not represent city populations.

test has not been repeated for the full ten-site calculation with its image-derived materials and exact specular term. The test validates this component. It does not provide external validation of every city result. The level-2 body coupling uses one-sided local incidence. It does not trace body self-occlusion.

The ten routes are selected case studies. Their differences do not rank the ten cities and do not estimate population exposure. Each result is normalized per unit $\rho_A P_{\mathrm{EIRP}}$, so it is not an absolute prediction for an operator deployment. The study uses one 15 GHz frequency, one body phantom, one body orientation that faces along the walk, one roofline transmitter model, and ten plaza routes. Additional frequencies, body models, headings, deployment laws, street canyons, parks, and repeated route selections are outside this study.

Image-derived materials cover only surfaces that an accepted camera can see and project onto the city mesh. Unlabeled material-map cells use the geometry-based material. Transient pixels do not enter the static surface. Image labels can preserve within-triangle material variation, but it cannot recover geometry that is absent from the photogrammetric mesh. The two-site material control tests how these assignments change the final result, but it supplies no image-label ground truth. Coverage over the complete city mesh also differs from coverage on the surfaces reached by propagation paths. The path audit reports where the recorded image-derived and geometry-based materials act in one-reflection specular and first-diffuse transport. It does not test whether the image labels or geometry-based material are correct. The current claims therefore apply to these recorded surfaces.